



GROUP TECHNICAL SOLUTIONS (GTS)

PETRONAS CHEMICALS LDPE SDN BHD

Performance Test Run

September 2021

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DCS CONFIGURATION FOR APC IMPLEMENTATION		

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1.0 INTRODUCTION

PETRONAS Chemicals LDPE Sdn Bhd (PCLDPE) has embarked on a project to revamp its Advanced Process Control application. This involves reviewing design of the advanced controller, inferential and the grade transition programme.

This document describes the modification and changes done on Grade Transition Manager (GTM)

2.0 OBJECTIVE

The objectives of this document are:

- To document all changes made during APC revamp on GTM
- To elaborate and explain the changes made on GTM

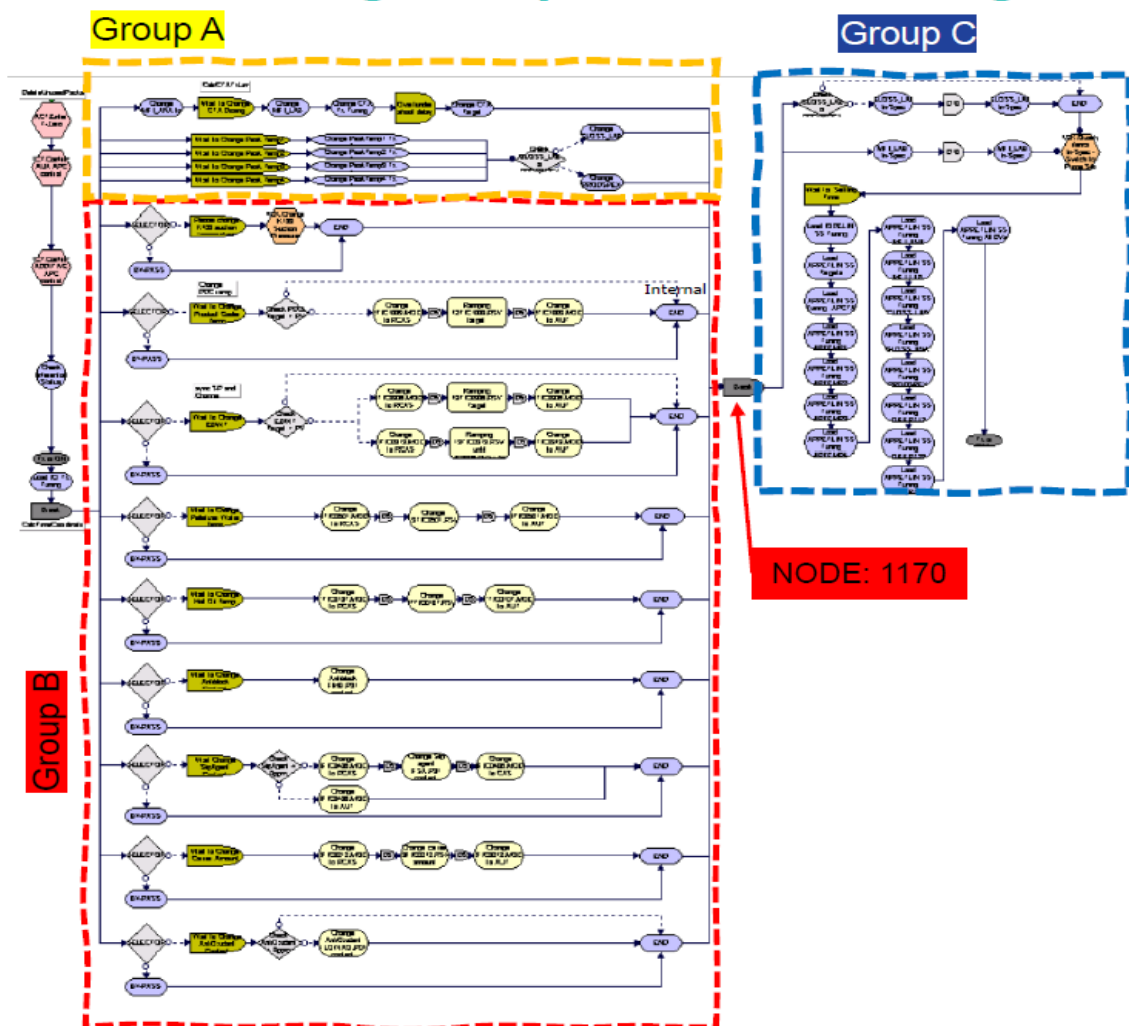
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3.0 GTM OVERVIEW

The Grade Transition Manager (GTM) product is used to implement the values configured in the recipe database. The recipe values are implemented according to the transition logic. Transitions will be a combination of closed-loop (automated) and logic that will require operator intervention. GTM will be able to implement values for APC related variables and other controllers(auxiliary).

The Process Recipe database holds the required recipe data as well as all transition logic. The transition logic consists of information required for working with Apollo to move the process from the product currently being made to the desired product

Original structure:



Above is the original GTM structure. To ease the process of rearrangement, the nodes has been grouped as below:

Group A: APC variables and target

Group B: Auxiliary target

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Group C: Quality Final Check and APC controller tuning load.

Generally, the new operator page is similar with old operator page but with a few added features and readings. Below is the summary.

1) No RST WDTIME button.

Operator will no longer required to click RST WDTIME button when starting APC. The watchdog is automatically started when APC is ON.

2) Off APC and MV button

Operator will be able to turn off any MV or APC from this operator page. However, if operator need to turn APC ON, operator would need to access client PC and turn ON from PCWS after doing some parameters checking.

3) APC-DCS Page link

There is a button to link APC Operator page to the main reactor page. This to ease operator navigation while operating the APC and plant.

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4.0 NEW TAG CREATION

A few new DCS tags had been created to be the inputs for APC in order to ensure the functionality of the new APC

The new DCS tags are per below:

No	DCS Tags	Description	Remarks
1	DTALLZONE.PV	(12TY0710.PV - 12TIC0102A.PV) + (12TY0810.PV 12TI0225A.PV) + (12TY0910.PV 12TI0301A.PV) + (12TI0940.PV 12TI0319A.PV)	The value is calculated by using CALCU block
2	CTARATIO.PV	11FIC0602.PV / 39FI0102B.PV	The value is calculated by using CALCU block
3	APPURGE	New PID creation. To be a dummy controller for APC remote setpoint.	
4	PRODSPEX_PVFLT.PV	Filtered value of Prodspeex	NUM= 5 / SMPL = 60
5	13PI0107B_PVFLT.PV	Filtered value of 13PI0107B	NUM= 5 / SMPL = 60
6	12PIC0103_PVFLT.PV	Filtered value of 12PIC0103	NUM= 5 / SMPL = 60
7	12TIC0102A_PVFLT.PV	Filtered value of 12TIC0102A	NUM= 5 / SMPL = 60
8	12TIC0401_PVFLT.PV	Filtered value of 12TIC0401	NUM= 5 / SMPL = 60
9	DTALLZONE_PVFLT.PV	Filtered value of DTALLZONE	NUM= 5 / SMPL = 60
10	11TI0302_PVFLT.PV	Filtered value of 11TI0302	NUM= 5 / SMPL = 60
11	CTARATIO_PVFLT.PV	Filtered value of CTARATIO	NUM= 5 / SMPL = 60
12	13TI0116B_PVFLT.PV	Filtered value of 13TI0116B	NUM= 5 / SMPL = 60
13	13JI0101_PVFLT.PV	Filtered value of 13JI0101	NUM= 5 / SMPL = 60

DTALLZONE Calculation Details:

LDPE\FCS0102\DR0054\DTALLZONE.edf	
General-Purpose Calculation Formula (DTALLZONE)	
Line Number	General-Purpose Calculation Formula
001	ALIAS T1 12TY0710.PV
002	ALIAS T2 12TIC0102A.PV
003	ALIAS T3 12TY0810.PV
004	ALIAS T4 12TI0225A.PV
005	ALIAS T5 12TY0910.PV
006	ALIAS T6 12TI0301A.PV
007	ALIAS T7 12TY0940.PV
008	ALIAS T8 12TI0319A.PV
009	
010	P01 = T1
011	P02 = T2
012	P03 = T3
013	P04 = T4
014	P05 = T5
015	P06 = T6
016	P07 = T7
017	P08 = T8
018	
019	CPV = ((T1-T2)+(T3-T4)+(T5-T6)+(T7-T8))

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CTARATIO Calculation Details

LDPE\FCS0102\DR0054\CTARATIO.edf	
General-Purpose Calculation Formula (CTARATIO)	
Line Number	General-Purpose Calculation Formula
001	ALIAS F2 39FI0102B.PV
002	
003	P01 = RV1
004	P02 = F2
005	
006	CPV = (RV1/F2)

Filtered variables details

The filter tags are created in order to introduce filter for APC inputs tags. Previously, the filtering has been done on the APC itself. However, it utilized APC tokens. Thus, in order to optimize APC tokens utilization, all filterings are configured on the DCS.

The filter block is using Moving Average Block. Below are the details of Moving Average block.

■ Set Parameters

The parameters of the Moving-Average Block (AVE-M) are shown as follows.

- Gain (GAIN):
A numeric value of 7 digits or less including the sign and decimal point.
- Number of Sample Point (NUM):
A numeric value between 1 and 60.
- Sampling interval (SMPL):
A numeric value between 0.1 and 10000.0. Unit: sec.

For the sampling interval (SMPL), set a multiple of the scan period. If any other value is set, sampling is performed at an interval rounded up to the next larger multiple of the scan period.

How the sampling interval (SMPL) value is rounded up is shown as follows:

Scan period = 1 second	SMPL = 0.5	→	Action occurs assuming SMPL = 1
Scan period = 1 second	SMPL = 1.1	→	Action occurs assuming SMPL = 2
Scan period = 1 second	SMPL = 2	→	Action at SMPL = 2
Scan period = 0.1 second	SMPL = 0.5	→	Action at SMPL = 0.5

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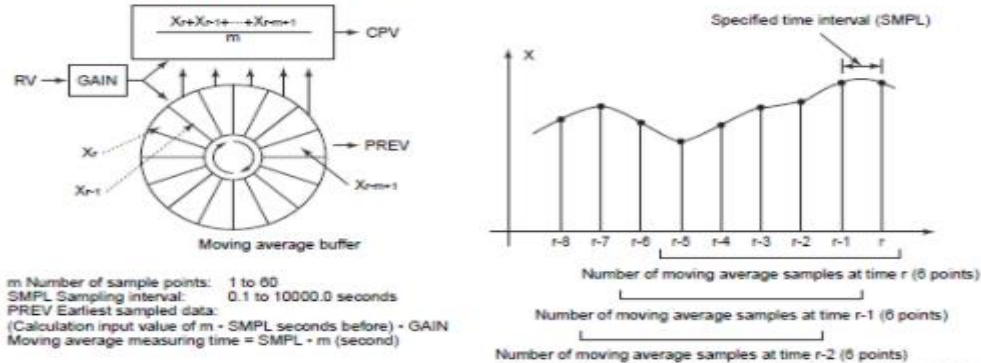
The Moving-Average Block (AVE-M) performs the following calculation processing to obtain the moving average of input data.

$$CPV = GAIN \cdot \frac{X_r + X_{r-1} + \dots + X_{r-m+1}}{m}$$

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The Moving-Average Block (AVE-M) calculates the average value of past data that have been sampled at a specified interval. The target input signals of this calculation are the specified number of latest sampled data.

The following shows an action example of the Moving-Average Block (AVE-M) when the specified number of samples is "6."



Initialization of the moving average is done by the reset switch (RST). When the reset switch (RST) is turned to "1," the buffer is initialized with the calculated input value (RV). When the initialization is complete, the reset switch (RST) returns to "0" (normal state). When the data status of the calculated output value (CPV) returns to normal from IOP+ (input open high) or CAL (calibration), the reset switch (RST) changes to "1" automatically and the moving average is initialized.

The earliest sampled data is stored in the earliest calculation input value (PREV) and is available for reference.

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5.0 DCS LOGIC

There are few logics have been created to ensure smooth APC operation. Below are the new added logics.

1) AWC(Anti Wind Up Calculation) Logic

LDPE\FCS0101\DR0027\APPURGE_AWC.edf	
General-Purpose Calculation Formula (APPURGE_AWC	
Line Number	General-Purpose Calculation Formula
001	PROGRAM
002	
003	IF ({11FIC0401.ALPM.MHI}) THEN
004	CPV=2
005	ELSE IF ({11FIC0401.ALPM.MLO}) THEN
006	CPV=1
007	ELSE IF ({11FIC0401.ALPM.NR}) THEN
008	CPV=0
009	END IF
010	
011	END

This logic to give information to APC whenever the base layer controller hit the saturation whether high or low saturation limit. This information will be used by APC to prevent the set point from increasing or decreasing further when the base layer controller is limited to constraint.

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2) Shed Mode logic

This sequence table serves as mode switch for base layer PID. If APC request is ON, base layer will be switched to CAS. Whenever the APC is OFF, this logic will shed MV mode from CAS to AUT.

[illegible]

3) Loop status check

This sequence table is to switch mode for APC PID. If APC request is ON, APPURGE will be switched to RCAS and Loop status switch will be ON. Otherwise, shed mode and service off request from DCS will be initiated.

LD01PC00101\39002\APPURGE_STI.edf - Edit Sequence Tables.edf

Sequence Tables (APPURGE_STI) 1

Start Timing		Periodic Execution Type(T)		Output Timing		Output Only When Conditions Change(C)		Scan Period		Basic Scan		Control Period		Control Phase		
NO.	Tag Name Data Store	Data	Function	U	L	1	2	4	5	6	7	8	9	10	11	12
			Function													
			Sequence Signal Comment													
C01	APC01001.DV	ON														
C02	APC01002.DV	ON														
C03	APC01003.DV	ON														
C04																
C05	APC01005.HISE	NCAS														
C06	APC01006.HV	0	APC01006.HV													
C07																
C08																
C09																
C10																
C11																
C12																
C13																
C14																
C15																
C16																
C17																
C18																
C19																
C20																
C21																
C22																
C23																
C24																
C25																
C26																
C27																
C28																
C29																
C30																
C31																
C32																
A01	APC01005.HISE	APC														
A02	APC01006.HISE	NCAS														
A03	APC01006.HV	0	APC01006.HV													
A04			APC01006.HV													
A05	APC01006.HV	0	APC01006.HV													
A06																
A07																
A08																
A09																
A10																

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6.0 YOKOGAWA APC- DCS CONFIGURATION REPORT

Below is the report produced by Yokogawa on the APC-DCS configuration report. Please refer to this document to see more details on the control drawing and logic details